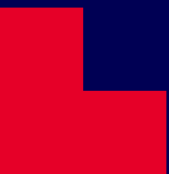

Artificial Intelligence

Week 5: First-order Logic (FOL)

COSC2129/2723/3118



Material

- Chapter 7, Book: Artificial Intelligence: A Modern Approach Global edition
- Book - TOC of Artificial Intelligence: A Modern Approach, 4th Global ed.
- AIMA: Pseudo code
- Good practices Python

Propositional logic: Limitation

- Example 1: Represent in PL
 - The table has three legs
 - The table is made of wood and glass
 - The table is expensive

➤ Properties of ONE object
- Example 2
 - Assignment 1 is challenging
 - Assignment 2 is challenging

➤ The same property of difference objects
- Example 3
 - Table 2 is next to Table 1
 - Table 2 is smaller than Table 1
 - The bag is on Table 2

• ?

Propositional logic (PL): Limitation

- KB:
 - All patients need treatment.
 - If someone is hospitalized, they are a patient.
 - John is in the hospital.
- Rules:
 - If $P \rightarrow Q$ & P , then deduce Q (Modus ponens)
 - From $\forall x P(x)$ we can conclude $P(a)$ for any individual a (Universal Instantiation).
- Conclusion:
 - John needs a treatment.
- Solution?

First-order logic

- First-order logic (FOL):
 - built on the foundation of PL
 - more expressive than PL
 - useful representational ideas (objects, relations, functions) from nature language.
 - **Objects**: Table, wumpus, people, houses, numbers, colors,...
 - **Relations/properties**: next, red, round, prime, brother of, smaller than, bigger than, part of, between, ...
 - **Functions**: father of, best friend, one more than, plus,...

First-order logic

- First-order logic (FOL) models the world in terms of
 - **Objects** are things with individual identities
 - **Properties** of objects that distinguish them from other objects
 - **Relations** that hold among sets of objects
 - **Functions** are a subset of relations where there is only one “value” for any given “input”
- Examples:
 - Objects: Students, lectures, companies, cars, ...
 - Relations: Brother-of, bigger-than, outside, part-of, has-color, occurs-after, owns, visits, precedes, ...
 - Properties: blue, wood, oval, even, large, ...
 - Functions: father-of, best-friend, second-half, one-more-than, ...

FOL: Syntax

Basic syntactic elements of FOL:

- **Constant symbols:** represent individuals in the world
 - Daisy (name)
 - 3 (legs)
 - Wood
- **Predicate symbols:** map individuals to truth values
 - greater($3 > 2$)
 - blue(Sky)
 - color(Grass, Green)
- **Function symbols:** map individuals to individuals
 - $\text{Sqrt}(4) = 2$
 - father-of(Daisy) = John
 - color-of(Sky) = Blue
 - legOfTable

FOL: Syntax

- Variable symbols
 - E.g., $x, y, \dots$
- Connectives
 - Same as in PL: not ($\neg$), and ($\wedge$), or ($\vee$), implies ($\rightarrow$), if and only if (biconditional $\leftrightarrow$)
- Quantifiers
 - Universal $\forall x$ or ($\mathbf{Ax}$)
 - Existential $\exists x$ or ($\mathbf{Ex}$)
- Equality: $=$

FOL: Sentences

Sentences are built from terms and atoms.

- **A term** (a real-world individual) is a constant, a variable, or an n-place function of n terms.
 - x and $f(x_1, \dots, x_n)$ are terms, where each x_i is a term.
 - A term with no variables is a **ground term**
- **An atomic sentence** (which has value true or false) is an n-place predicate of n terms
 - $\text{broken}(\text{Table1}, \text{Table2})$ (term: objects, fact)
 - $\text{mother}(\text{Jody}, \text{Tim})$ (predicate: relation, fact)
- **A complex sentence** is formed from atomic sentences connected by the logical connectives:
 - $\neg P, P \vee Q, P \wedge Q \vee R, P \rightarrow Q, P \leftrightarrow Q$ where P, Q, R are sentences

FOL: Sentences (con't)

- A quantified sentence adds quantifiers $\forall$ and $\exists$
- A well-formed formula (wff) is a sentence containing no “free” variables. That is, all variables are “bound” by universal or existential quantifiers.

$(\forall x)P(x,y)$ has x bound as a universally quantified variable, but y is free.

Examples:

Somebody likes Jane: $\exists x : (\text{likes}(x, \text{Jane}))$.

Somebody likes everybody: $\exists x : (\forall y : (\text{likes}(x, y)))$

$\forall x : (\text{even}(x) \wedge \text{prime}(x) \Rightarrow (x = 2))$ (a mathematical fact)

Examples

- See slides: Lecture 05

Quantifiers

- **Universal** quantification
 - $(\forall x)P(x)$ means that P holds for **all** values of x in the domain associated with that variable
 - E.g., $(\forall x) \text{dolphin}(x) \rightarrow \text{mammal}(x)$
- **Existential** quantification
 - $(\exists x)P(x)$ means that P holds for **some** value of x in the domain associated with that variable
 - E.g., $(\exists x) \text{study}(x, \text{AI}) \wedge \text{smart}(x)$

Quantifiers

- Universal quantifiers: often used with “implies” to form “rules”:
 $(\forall \mathbf{x}) \text{ student}(\mathbf{x}) \rightarrow \text{smart}(\mathbf{x})$ means “All students are smart”
 $(\forall \mathbf{x}) \text{ student}(\mathbf{x}) \wedge \text{smart}(\mathbf{x})$ means “Everyone in the world is a student and is smart”
- Existential quantifiers: usually used with “and” to specify a list of properties about an individual:
 $(\exists \mathbf{x}) \text{ student}(\mathbf{x}) \wedge \text{smart}(\mathbf{x})$ means “There is a student who is smart”
- A common mistake is to represent this English sentence as the FOL sentence:
 $(\exists \mathbf{x}) \text{ student}(\mathbf{x}) \rightarrow \text{smart}(\mathbf{x})$
 - But what happens when there is a person who is *not* a student?

Multiple quantifiers (Scope)

- Switching the order of universal quantifiers *does not* change the meaning:
 - $(\forall x)(\forall y)P(x,y) \leftrightarrow (\forall y)(\forall x) P(x,y)$
- Similarly, you can switch the order of existential quantifiers:
 - $(\exists x)(\exists y)P(x,y) \leftrightarrow (\exists y)(\exists x) P(x,y)$
- Switching the order of universals and existentials *does* change meaning:
 - Everyone likes someone: $(\forall x)(\exists y) \text{ likes}(x,y)$
 - Someone is liked by everyone: $(\exists y)(\forall x) \text{ likes}(x,y)$

Connections between All and Exist

We can relate sentences involving $\forall$ and $\exists$ using De Morgan's laws:

$$(\forall x) \neg P(x) \leftrightarrow \neg(\exists x) P(x)$$

$$\neg(\forall x) P \leftrightarrow (\exists x) \neg P(x)$$

$$(\forall x) P(x) \leftrightarrow \neg (\exists x) \neg P(x)$$

$$(\exists x) P(x) \leftrightarrow \neg(\forall x) \neg P(x)$$

Knowledge Representation in FOL

- A domain: a specific area of the world that we aim to represent knowledge about.
- A set of **constants, predicates and functions** needs to be determined to fit the domain.

We assume they have been determined for the examples of predicates/functions.

Semantics of FOL

- **Domain M**: the set of all objects in the world (of interest)
- **Interpretation**: includes
 - Assign each *constant* to an object in M
 - Define *functions* (a mapping $M^n \Rightarrow M$)
 - Define *predicates* (a mapping $M^n \Rightarrow \{T, F\}$)
 - Every ground predicate with any instantiation will have a truth value
 - In general there is an infinite number of interpretations because $|M|$ is infinite
- **Define logical connectives**: $\sim, \wedge, \vee, \Rightarrow, \Leftrightarrow$ as in PL
- **Define semantics of $(\forall x)$ and $(\exists x)$**
 - $(\forall x) P(x)$ is true iff $P(x)$ is true under all interpretations
 - $(\exists x) P(x)$ is true iff $P(x)$ is true under some interpretation

Semantics of FOL

- **Model**: an interpretation of a set of sentences such that every sentence is *True*
- A sentence is
 - **satisfiable** if it is true under some interpretation
 - **valid** if it is true under all possible interpretations
 - **inconsistent** if there does not exist any interpretation under which the sentence is true
- **Logical consequence**
 $S \models X$ if all models of S are also models of X

Example (1)

- All students know AI

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- All students know AI.
 $\forall x \text{ Student}(x) \wedge \text{Knows}(x, \text{AI})$

Example (2)

- There is some course that every student has taken

Example (2)

- There is some course that every student has taken
 $\exists y \text{ Course}(y) \wedge (\forall x \text{ Student}(x) \rightarrow \text{Taken}(x,y))$

Example (3)

American law says that it is a crime for Americans to sell weapons to hostile nations. The country Nono, an enemy of America, has some missiles, and all of its missiles were sold to it by Colonel West, who is an American. Prove that Colonel West is a criminal.

- Approach:
 - What are objects?
 - Constants?
 - Relationships?
 - Properties?

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- Approach:
 - What are objects?
 - Constants?
 - Relationships?
 - Properties?
 - Decide predicates and functions
 - Construct a **knowledge base**

Example (3)

Based on the story and common sense, a knowledge base can be set as:

1. $\forall x, y, z \text{ American}(x) \wedge \text{Weapon}(y) \wedge \text{Sells}(x, y, z) \wedge \text{Hostile}(z) \Rightarrow \text{Criminal}(x)$
... it is a crime for an American to sell weapons to hostile nations:
2. $\text{Owns}(\text{Nono}, \text{M1})$
3. $\text{Missile}(\text{M1})$
Nono ... has some missiles,
4. $\forall x \text{ Missile}(x) \wedge \text{Owns}(\text{Nono}, x) \Rightarrow \text{Sells}(\text{West}, x, \text{Nono})$
... all of its missiles were sold to it by Colonel West
5. $\forall x \text{ Missile}(x) \Rightarrow \text{Weapon}(x)$
Missiles are weapons
6. $\forall x \text{ Enemy}(x, \text{America}) \Rightarrow \text{Hostile}(x)$
An enemy of America counts as "hostile"
7. $\text{American}(\text{West})$
West, who is American ...
8. $\text{Enemy}(\text{Nono}, \text{America})$
The country Nono, an enemy of America ...

Example (3)

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Inference in FOL

- Inference in FOL is an extension of inference in PL.

Quantified inference rules

- Universal instantiation
 - $\forall x P(x) \therefore P(A)$
- Universal generalization
 - $P(A) \wedge P(B) \dots \therefore \forall x P(x)$
- Existential instantiation
 - $\exists x P(x) \therefore P(F)$ ← skolem constant F
- Existential generalization
 - $P(A) \therefore \exists x P(x)$

Universal instantiation (universal elimination)

- If $(\forall x) P(x)$ is true, then $P(C)$ is true, where C is *any* constant in the domain of x
- Example:
$$(\forall x) \text{learn}(\text{Man}, x) \Rightarrow \text{learn}(\text{Man}, \text{painting})$$
- The variable symbol can be replaced by any ground term (any constant symbol or function symbol applied to ground terms only).

Existential instantiation (existential elimination)

- From $(\exists x) P(x)$ infer $P(c)$
- Example:
 $(\exists x) \text{learn}(\text{Man}, x) \rightarrow \text{learn}(\text{Man}, \text{Subject})$
- Note that the variable is replaced by a **brand-new constant** not occurring in this or any other sentence in the KB
- Also known as skolemization; constant is a **skolem constant**
- Convenient to use this to reason about the unknown object, rather than constantly manipulating the existential quantifier.

Existential generalization (existential introduction)

- If $P(c)$ is true, then $(\exists x) P(x)$ is inferred.
- Example
 $\text{learn}(\text{Man}, \text{Painting}) \Rightarrow (\exists x) \text{learn}(\text{Man}, x)$
- All instances of the given constant symbol are replaced by the new variable symbol
- Note that the variable symbol cannot already exist anywhere in the expression.

Propositionalization (FOL $\rightarrow$ PL)

- Inference in FOL is an extension of inference in PL.
- We can **reduce** FOL inference to PL inference via propositionalization.
(Need to deal with *substitutions* and *quantifiers*)
- The basic idea
 - An existential quantifier can be substituted by one instantiation.
 - A universal quantifier can be substituted by the set of all possible instantiations

Generalized Modus Ponens (GMP)

- All variables assumed universally quantified

Modus Ponens can be extended (lifted) for inference in FOL – **Generalized Modus Ponens**.

$$\frac{p'_1, p'_2, \dots, p'_n, (p_1 \wedge p_2 \wedge \dots \wedge p_n) \Rightarrow q}{SUBST(\theta, q)}$$

where θ is a substitution such that for all i , $SUBST(\theta, p'_i) = SUBST(\theta, p_i)$

There are $n+1$ premises to this rule: the n atomic sentences p'_i and the one implication.

It is basically modus ponens + to find a substitution for the variables in sentences.

GMP Rule Format

- Remember Modus ponens: If $P \rightarrow Q$ & P are true, then Q is true.
- GMP:
 - Given:

$$\forall x_1, x_2, \dots, x_n (P_1(x) \wedge P_2(x) \wedge \dots \wedge P_k(x)) \rightarrow Q(x)$$

- And

$$P_1(c), P_2(c), \dots, P_k(c)$$

- Then you can infer

$$Q(c)$$

GMP example

- All variables assumed universally quantified

$$\frac{p_1', p_2', \dots, p_n', (p_1 \wedge p_2 \wedge \dots \wedge p_n \Rightarrow q)}{q\theta} \quad \text{where } p_i'\theta = p_i\theta \text{ for all } i$$

- Rule: $King(x) \wedge Greedy(x) \Longrightarrow Evil(x)$ ■
 - Precondition of rule: p_1 is $King(x)$ p_2 is $Greedy(x)$
 - Implication: q is $Evil(x)$ ■
 - Facts: p_1' is $King(John)$ p_2' is $Greedy(y)$ ■
 - Substitution: θ is $\{x/John, y/John\}$
- $\Rightarrow$ Result of modus ponens: $q\theta$ is $Evil(John)$

Unification

- The process of **finding substitutions**
- A unification algorithm takes two sentences and returns a unifier (substitution) for them if one exists.

Examples:

	Sentence 1	Sentence 2	θ
1	$p(\text{John})$	$p(x)$	$\{x/\text{John}\}$
2	$p(\text{John})$	$p(\text{Richard})$	Failure
3	$p(\text{John}, x)$	$p(y, \text{Mary})$	$\{y/\text{John}, x/\text{Mary}\}$
4	$p(x, x)$	$p(\text{John}, y)$	$\{x/\text{John}, y/\text{John}\}$
5	$p(x, x)$	$p(\text{John}, \text{Richard})$	Failure
6	$p(x)$	$p(y)$	$\{x/y\}$

Most general unifier (MGU)

- Unification should only return the most general one, which makes the least commitment to constants.
- The algorithm for computing MGUs can be found in the textbook.

Example of Inference in FOL

- Prove that Colonel West is a criminal.
- See slides.

Question?